Notes on "A function related to the Mordell-Weil rank of Elliptic Curves" by Prof. Siddhi Pathak

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1 Why I'm reading this

1.1 How I found your paper

I was looking for a reading project on a topic I would enjoy working on. Analysis was the first choice. As I skimmed through CMI professors' papers, I found yours and I found it accessible enough that I could devote some time and make some sense of it.

1.2 How long it took and why?

It took me a little more than a week. Most of my time was spent on understanding the machinery – as one can see by the questions that arose on my first read. The proofs of some of the Lemmas and the main theorem, were easy enough to grasp once the unproven Lemmas were considered as black boxes – although I do plan on learning their proofs soon.

2 Roadmap of the Paper

- The paper presents the main theorem in the introduction, claiming that $f_p(n)$ is unbounded as n ranges through squarefree positive integers, right after defining $f_p(n)$.
- Section 2 is mostly concerned with the descent strategy showing that it has applications in Algebraic Geometry i.e., by stating and sort of proving through already established results, that the Mordell-Weil rank of elliptic curves is bounded below by the cognate function $f(n)$. This motivates us to actually consider the analytic nature of the function $f_p(n)$. More specifically, whether $f_p(n)$ is bounded or not.
- Section 3 establishes 4 lemmas, each of which hints at how $x \log x$ crops up in the final equation that is to be proved.
- Finally, Section 4 proves the main theorem with the results acquired so far.

3 Important results

For me, the biggest hurdles while trying to understand Lemma 5 was: knowing which quantity was fixed, how the inner sums were swapped with outer sums.

Theorem 1 *(Main) The main theorem states that $f_p(n)$ is unbounded for squarefree integers n and p is prime; $f_p(n)$ is defined as follows:*

$$f_p(n) := \#\{1 \leq a, b \leq n : ab = n, \left(\frac{a+b}{p}\right) = 1\}$$

To be more precise, it states that

$$\sum_{\substack{n \leq x \\ n \text{ squarefree}}} f_p(n) \gg x \log x$$

Equivalently,

$$\sum_{\substack{n\leq x\\ n\ squarefree}} f_p(n) \geq Cx\log x$$

for $C = C(p)$, a constant that depends on p.

The motivation behind considering $f_p(n)$ is the more general function $f(n)$. It has been shown that if $f(n)$ is unbounded, then there are elliptic curves over the rational number field with arbitrarily large rank. Moreover, it has been shown by David Clark, that $f(n)$ is unbounded if we drop the condition that n must be squarefree. This paper studies unboundedness of $f_p(n)$ when n is squarefree to get further along in proving that $f(n)$ is also unbounded under the same conditions.

Lemma 1 *(Tauberian theorem) Let $f(s)$ be a Dirichlet function, $f(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$ with $a_n = O(n^\epsilon)$ which is another way of saying that $a_n \in (-C_\epsilon n^\epsilon, C_\epsilon n^\epsilon)$ for a fixed $\epsilon > 0$. Suppose that f is factorizable into two functions*

$$f(s) = \zeta(s)^k g(s)$$

for a natural number k, such that $g(s)$ is also a Dirichlet function with the special condition that it is absolutely convergent in $\Re(s) > 1-\delta$ for some $0 < \delta < 1$. Then, the finite sum $\sum_{n\leq x} a_n$ is asymptotically equivalent to a constant multiple of $x(\log x)^{k-1}$. To put it more precisely,

$$\sum_{n\leq x} a_n \sim \frac{g(1)}{(k-1)!} x(\log x)^{k-1}$$

as $x \to \infty$. In essence, to find a "close enough" approximation for the sum of an infinite series, we can construct the function f and invoke the Tauberian theorem. The structure itself hasn't revealed itself to me and I'm considering this a black-box for now.

Lemma 2 *We define $R(x; a, q)$ to be the number of squarefree integers in the arithmetic progression $a \pmod q$ with $(a, q) = 1$. For more, refer to the relevant question on "Things I didn't understand on my first read". For a simple example, consider $q = 5, a = 2$ and $x = 12.5$. Then,*

$$2, 7, 12, \ldots$$

has two squarefree terms less than $x = 12.5$ (12 is not square-free, as $4 = 2^2 \mid 12$). Thus,$R(12.5; 2, 5) = 2$ We give a rather close approximation to this function by writing it as the sum of a main term and a controlled error term:

$$R(x; a, q) = \frac{1}{\zeta(2)} \prod_{\substack{p\mid q\\ p\ is\ prime}} \left(1 - \frac{1}{p^2}\right)^{-1} \frac{x}{q} + O_\epsilon\left((\frac{x}{q})^{1/2} + q^{1/2+\epsilon}\right)$$

In order to not concern myself with the deep trenches of the proof behind this absolute monstrosity, I am treating this as a black box. However, one may connect this to the function $R(x)$, which is asymptotically equivalent to $\frac{x}{\zeta(2)}$ in the sense that both $R(x)$ and $R(x; a, q)$ have their main terms linear in x. They are similar in the way that both of them count the number of squarefree integers bounded by x (with the added constraints for $R(x; a, q)$)

Lemma 3 *We will now state a result which shows that the sum of divisors of squarefree integers n each bounded by a real number $x \in \mathbb{R}$ is comparable to, or can be approximated by, a constant multiple of $x \log x$. Notice that we had used $x \log x$ in Lemma 1 too, in the expression $x(\log x)^{k-1}$.*

To sift out the squarefree integers, we use the Mobius function

$$\sum_{n\leq x} \mu^2(n)d(n) = Cx\log x + o(x\log x)$$

where once again the main term is $Cx \log x$ and the error term is $o(x \log x)$. The constant C can be derived to be positive and non-zero. More precisely,

$$C = \prod_p \left(1 - \frac{3}{p^2} + \frac{2}{p^3}\right) = 0.33.. > 0$$

Lemma 4 *For a prime p,*

$$\left| \sum_{\substack{ab \le x \\ a,b \ squarefree}} \left(\frac{a+b}{p}\right) \right| \le \frac{1}{\sqrt{p}(p+1)\zeta(2)^2} x \log x + O(x)$$

Here too we see that the sum is close to a constant multiple of $x \log x$.

4 Main ideas of the proofs

Proofs of Lemmas 1 and 2 are omitted for the sake of brevity.

Proof of Lemma 3:
We use Lemma 1 – Take $f(s) = \sum_{n=1}^{\infty} \frac{u^2(n)d(n)}{n^s}$ with $a_n = \mu^2(n)d(n)$. Before we factorize the function, we will use this fact:

$$f(s) = \prod_p \left(1 + \frac{2}{p^s}\right)$$

Since $\mu^2(n)$ and $d(n)$ is multiplicative, so is $a_n = a(n) = \mu^2(n)d(n)$. We know that by Fundamental Theorem of Arithmetic, $n = \prod_{p|n} p^{v_p(n)}$. Hence,

$$\frac{1}{n^s} = \frac{1}{\prod_{p|n} p^{v_p(n)s}}$$

and

$$a(n) = \prod_{p|n} a(p^{v_p(n)})$$

Therefore,

$$\frac{a_n}{n^s} = \prod_p \frac{a(p^{v_p(n)})}{p^{v_p(n)s}}$$

Summing the LHS over all natural numbers $n = 1, 2, \ldots$, we get

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s} = f(s) = \prod_p \left(\sum_{k=0}^{\infty} \frac{a(p^k)}{p^{ks}} \right)$$

We know that

$$\mu^2(n) = \begin{cases} 1 & \text{n is square-free or n=1} \\ 0 & \text{n is divisible by the square of some prime} \end{cases}$$

Thus, whenever $n = 1$, or $n = p$, $\mu^2(n) = 1$; otherwise, $\mu^2(n) = 0$. Therefore, we need to consider only the cases where $k = 0, 1$ for the sum inside the product for $f(s)$. We have $d(1) = 1$, $d(p) = 2$. Thus,

$$\boxed{f(s) = \prod_p \left(\frac{a(1)}{1} + \frac{a(p)}{p^s} \right) = \prod_p \left(1 + \frac{2}{p^s}\right)}$$

Now, we factorize the product as follows:

$$\prod_p \left(1 + \frac{2}{p^s}\right) = \prod_p \left(1 - \frac{1}{p^s}\right)^{-2} \prod_p \left(1 - \frac{1}{p^s}\right)^2 \left(1 + \frac{2}{p^s}\right) = \zeta(s)^2 g(s)$$

where

$$\zeta(s)^2 = \prod_p \left(1 - \frac{1}{p^s}\right)^{-2}$$

and

$$g(s) = \prod_p \left(1 - \frac{1}{p^s}\right)^2 \left(1 + \frac{2}{p^s}\right)$$

Applying Lemma 2 gives us

$$\sum_{n \leq x} \mu^2(n) d(n) = Cx \log x + o(x \log x)$$

where

$$C = \frac{g(1)}{(2-1)!} = g(1) = \prod_p \left(1 - \frac{1}{p}\right)^2 \left(1 + \frac{2}{p}\right) = \prod_p \left(1 - \frac{3}{p^2} + \frac{2}{p^3}\right) = 0.33...$$

Proof Outline of Lemma 4:

- We begin by expressing the Legendre symbol using a Gauss sum. The reason for doing so is to exploit the property of exponential functions, $e^{a+b} = e^a e^b$.
- Then, the innermost sum

$$\sum_{\substack{ab \leq x \\ a,b \text{ squarefree}}} e\left(\frac{c(a+b)}{p}\right) = \sum_{\substack{a \leq x \\ a \text{ squarefree}}} e\left(\frac{ca}{p}\right) \sum_{\substack{b \leq \frac{x}{a} \\ b \text{ squarefree}}} e\left(\frac{cb}{p}\right)$$

  where the inner sum is calculated by fixing a squarefree integer $a \leq x$. After this we consider the expression

$$E(Y) = \sum_{\substack{b \leq Y \\ b \text{ squarefree}}} e\left(\frac{eb}{p}\right)$$

- To sift out the non-squarefree numbers, we use the Mobius function

$$\mu^2(b) = \sum_{t^2 | b} \mu(t)$$

  and get the expression involving double sums.

- We consider cases when p divides t and when it doesn't. Each contributing a factor to the sum. If $p \nmid t$, then the inner sum is bounded contributing $O(\sqrt{Y})$. If $p \mid t$, the contribution of $E(Y)$ is precisely $\sum_{t \leq \sqrt{Y} p | t} \mu(t) \lfloor \frac{Y}{t^2} \rfloor$.
- The sum obtained at last, is approximately the infinite sum

$$\sum_{\substack{t=1 \\ p|t}}^{\infty} \frac{\mu(t)}{t^2} = \frac{-1}{(p^2-1)\zeta(2)}$$

- Therefore,

$$\left| \sum_{t \leq \sqrt{Y} p | t} \mu(t) \lfloor \frac{Y}{t^2} \rfloor \right| \leq \frac{Y}{(p^2-1)\zeta(2)} + O(\sqrt{Y})$$

- Combining these with the facts: $|\tau| = \sqrt{p}$, and the estimates such as

$$R(x) = \frac{x}{\zeta(2)} + O(\sqrt{x}), \quad \sum_{\substack{n \leq x \\ n \text{ squarefree}}} \frac{1}{n} = \frac{\log x}{\zeta(2)} + O(1)$$

  we get the lemma.

There are many steps that are not partly/completely justified – I am treating those as black boxes for now, as the arguments are technically involved. For the purposes of this document, I focus only on understanding the structure rather than the entire machinery.

Proof of the main theorem:

We'll be splitting into two cases: when $p = 2$ and $p \geq 3$. For $p = 2$, the paper observes directly that $f_p(n) = f_2(n) = \#\{1 \leq a, b \leq n : ab = n\} = d(n)$. By Lemma 4, it has been established that

$$\sum_{\substack{n \leq x \\ \text{n squarefree}}} d(n) = Cx \log x + o(x \log x)$$

Therefore, the theorem holds true for $p = 2$.

For $p \geq 3$, we will be using the identity

$$2f_p(n) = \sum_{\substack{1 \leq a,b \leq n \\ \text{n squarefree}}} \left(\left(\frac{a+b}{p} \right) + 1 \right) - \sum_{\substack{1 \leq a,b \leq n \\ ab=n, p \mid a+b}} 1$$

the proof of which is quite beautiful as we'll witness now.
We'll work from $\left(\frac{a+b}{p}\right)$ to $f_p(n)$. By definition,

$$L = \left(\frac{a+b}{p} \right) = \begin{cases} 1 & a+b \text{ is a nonzero quadratic residue mod p} \\ -1 & a+b \text{ is a quadratic non-residue mod p} \\ 0 & p \mid a+b \end{cases}$$

We know that $f_p(n)$ only counts those $a + b$ for which $L = 1$. Thus, when trying to express $f_p(n)$ as a sum involving L, we must discard the contribution from the other cases. That is, we need to zero out each undesirable case. Consider $L + 1$:

$$L + 1 = \begin{cases} 2 \\ 0 \\ 1 \end{cases}$$

where $L + 1$ takes 2 when $a + b$ is a nonzero quadratic residue mod p, and the rest follows. Here, the unfavourable case is when $p \mid a + b$. We must simply remove this by considering the indicator function

$$1_{\{p \mid a+b\}} = \begin{cases} 1 & p \mid a+b \\ 0 & p \nmid a+b \end{cases}$$

Therefore, $L + 1 - 1_{\{p \mid a+b\}}$ is counted in $2f_p(n)$ (Note that the 2 comes from having to add 1, and sort of normalize the Legendre symbol). The function $f_p(n)$ is then

$$2f_p(n) = \sum_{ab=n} (L+1) - \sum_{ab=n} 1_{\{p \mid a+b\}} = \sum_{ab=n} \left(\left(\frac{a+b}{p} \right) + 1 \right) - \sum_{ab=n, p \mid a+b} 1$$

Let,

$$S(x) = 2 \sum_{\substack{n \leq x \\ \text{n squarefree}}} f_p(n)$$

Then,

$$S(x) = \sum_{\substack{n \leq x \\ \text{n squarefree}}} \sum_{ab=n} \left(\frac{a+b}{p} \right) + \sum_{\substack{n \leq x \\ \text{n squarefree}}} \sum_{ab=n} 1 - \sum_{\substack{n \leq x \\ \text{n squarefree}}} \sum_{ab=n, p \mid a+b} 1$$

We will denote the 3 summations as S_1, S_2, and S_3 respectively. The goal is to get a lower bound for $S(x)$ involving the term $x \log x$. For that, we'll be finding an upper bound for S_3 and lower bounds for S_1, S_2.

- S_2 is estimated by Lemma 3:

$$\sum_{\substack{n\le x\\ \text{n squarefree}}}\sum_{ab=n}1=\sum_{\substack{n\le x\\ \text{n squarefree}}}d(n)=\sum_{n\le x}\mu^2(n)d(n)=Cx\log x+o(x\log x)\tag{1}$$

  where $C=0.33..$

- S_3 is estimated by Lemma 2: Since a,b are coprime,

$$S_3=\sum_{\substack{n\le x\\ \text{n squarefree}}}\sum_{ab=n,p|a+b}1=\sum_{\substack{a\le x\\ \text{a squarefree}}}\sum_{b\le\frac{x}{a},p|a+b\,\gcd(a,b)=1\text{b squarefree}}1\le\sum_{\substack{a\le x\\ \text{a squarefree}}}\sum_{b\le\frac{x}{a},p|a+b\text{b squarefree}}1$$

  For a,b in the inner sum, $a+b\equiv 0\pmod p\implies b\equiv -a\pmod p$. This allows us to invoke Lemma 2 with $\epsilon=\frac{1}{4}$ ($R(x)$ is almost linear). Using the estimates

$$\sum_{\substack{n\le x\\ n\text{ squarefree}}}\frac{1}{n}=\frac{\log x}{\zeta(2)}+O(1),\qquad\sum_{\substack{n\le x\\ n\text{ squarefree}}}\frac{1}{\sqrt{n}}=\frac{2\sqrt{x}}{\zeta(2)}+O(\log x).$$

  we can finally derive the upper bound for S_3 also:

$$S_3\le\frac{1}{\zeta(2)^2}\frac{p}{p^2-1}x\log x+O_p(x)$$

- S_1 is estimated by Lemma 4:

  Using the fact that $n=ab$ is squarefree $\iff\gcd(a,b)=1$, S_1 can be rewritten as

$$S_1=\sum_{\substack{n\le x\\ \text{n squarefree}}}\sum_{ab=n}\left(\frac{a+b}{p}\right)=\sum_{\substack{ab\le x\\ \text{a,b squarefree}\\ \gcd(a,b)=1}}\left(\frac{a+b}{p}\right)=\sum_{\substack{ab\le x\\ \text{a,b squarefree}}}\left(\frac{a+b}{p}\right)\sum_{d|a,d|b}\mu(d)$$

  as the inner sum equals exactly 1 if $\gcd(a,b)=1$, since $d|\gcd(a,b)$; Now, we can swap the sums to get

$$S_1=\sum_{d\le x}\mu(d)\sum_{\substack{ab\le x\\ \text{a,b squarefree}\\ d|a,d|b}}\left(\frac{a+b}{p}\right)$$

  Since $d|a$, and $d|b$, $a=da'$ and $b=db'$ for some integers a',b'. Then,

$$ab\le x\iff a'b'\le\frac{x}{d^2}$$

  Using the fact that the Legendre symbol is multiplicative,

$$\left(\frac{a+b}{p}\right)=\left(\frac{d(a'+b')}{p}\right)=\left(\frac{d}{p}\right)\left(\frac{a'+b'}{p}\right)$$

  Therefore,

$$S_1=\sum_{d\le x}\mu(d)\left(\frac{d}{p}\right)\sum_{\substack{ab\le x/d^2\\ \text{a,b squarefree}}}\left(\frac{a+b}{p}\right)$$

  Using Lemma 5 on the inner sum, we get, after some more algebraic manipulations:

$$|S_1|\le\frac{1}{\sqrt{p}(p+1)\zeta(2)}x\log x+O(x)$$

Now, we have everything we need to find the lower bound $S(x)$ which gives the lower bound for the summatory function of $f_p(n)$. Since $S(x)=S_1+S_2-S_3$, and using the 3 results (bounds) we derived,

$$S(x)\ge C'x\log x+o_p(x\log x)$$

The constant factor $C'=[C-\frac{1}{\sqrt{p}(p+1)\zeta(2)}-\frac{p}{(p^2-1)\zeta(2)^2}]$ is minimized when $p=3$; At $p=3$, $C'=0.10..>0$. Therefore, $S(x)\gg x\log x$ and consequently

$$\sum_{\substack{n\le x\\ \text{n squarefree}}}f_p(n)\gg x\log x$$

5 Things I didn't understand on my first read

- Elliptic curve
  Resolved: A smooth, projective curve over a field K of genus 1 with a K-rational point. What do all these mean though? Smooth – ensures that the curve has no crossings or cusps. Projective – Instead of working with affine space $\mathbb{R}^n$ we include points at infinity. (to put it roughly). A curve over a field K simply means that the equation is defined with coefficients in K. K-rational point is a point on the curve, say (x, y) such that $x, y \in K$. genus - it's an intrinsic geometric invariant that is closely associated with the topology of the curve. For $E(\mathbb{C})$, it has genus 1 since it's topologically a torus.

- Projective curve over field K
  Resolved: Consider points (x, y) in the affine space $\mathbb{R}^2$. In Euclidean space, two parallel lines can never intersect, by definition. Projective geometry roughly includes the case where even parallel lines may intersect at infinity. For this, we do a few things: instead of the usual coordinate (x, y) we consider the projective coordinates $[X : Y : Z]$ where it has a unique property

  $$[X : Y : Z] = [\lambda X : \lambda Y : \lambda Z]$$

  for $\lambda \neq 0$
  Whenever $Z \neq 0$, we can divide everything by Z to recover the original coordinates $x = X/Z$ and $y = Y/Z$. Whenever $Z = 0$, we can't divide by Z. These points $[X : Y : 0]$ are precisely the points at infinity, where not both X and Y are zero (keep in mind, for a K-rational projective point $[X : Y : 0], \quad X, Y \in K$)

  > Consider the curve $E : y^2 = x^3 + ax^2 + bx, \quad a, b \in \mathbb{Z}$. Set $x = X/Z$, $y = Y/Z$ to get
  >
  > $$Y^2Z = X^3 + aX^2Z + bXZ^2$$
  >
  > This is called the **projective closure** of E. To get points at infinity, set $Z = 0$:
  >
  > $$X = 0$$
  >
  > Therefore, the points at infinity are $[0, Y, 0]$; Dividing by Y we get $[0, 1, 0]$.

- Rank (or Mordell-Weil rank) of an elliptic curve
  Resolved: For an elliptic curve E over $\mathbb{Q}$, the Mordell-Weil theorem states that

  $$E(\mathbb{Q}) \cong E(\mathbb{Q})_{tors} \oplus \mathbb{Z}^r$$

  where $E(Q)_{tors}$ is the finite subgroup of torsion points. The integer r is called the rank of E.
  The rank measures the number of independent infinite-order rational points needed to generate the non-torsion part of $E(\mathbb{Q})$.

  > This is analogous to dimension/rank in linear algebra: just as the rank of a matrix measures the number of independent directions among its columns (dimension of column space), the rank of an elliptic curve measures the number of independent generators of the infinite part of its rational-point group (number of infinite-order rational points that can generate any other rational point. These independent rational points belong to the non-torsional subgroup of $E(\mathbb{Q})$)

- Legendre symbol
  Resolved: We define $\left(\frac{a}{p}\right)$ as

  $$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{a is a nonzero quadratic residue (mod p)} \\ -1 & \text{a is a quadratic non-residue (mod p)} \\ 0 & p|a \end{cases}$$

- $E(\mathbb{Q})$ notational meaning - point on this. order of an element (?)
  Resolved: $E(\mathbb{Q})$ is a set, moreover a group. P belongs to this set iff it is a point on the curve and $P \in \mathbb{Q}$. Order of P in this group is defined in the same group-theoretic way. For $G = E(Q)$, the operation is addition and so, the order of $P \in E(Q)$ is the smallest integer n such that

  $$nP = O$$

  where O is the identity element $[0 : 1 : 0]$ in $E(\mathbb{Q})$. If there exists no such n, then P has infinite order. Otherwise, it's an element of $E(\mathbb{Q})_{tors}$, i.e., the torsion subgroup of $E(\mathbb{Q})$

- Why does $W = (0, 0)$ have order 2 in group $E(\mathbb{Q})$?

  Resolved: We claim that for the group $E(\mathbb{Q})$, the additive inverse of a point $P(a, b)$ is given by $-P = -(a, b) = (a, -b)$: consider this a black box.
  Since $-W = -(0, 0) = (0, -0) = (0, 0) = W$, we have that $W + W = 2W = 0$. Thus, the order of $W = (0, 0)$ is 2 in this group.

- What does $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ mean? Can you generalise this and explain it through the general form?

  Resolved: For two arbitrary groups G and H, G/H denotes the quotient group, which is a collection of classes – elements of this set are not elements of G but rather the cosets of H i.e., gH for some $g \in G$. Consider two elements $a, b \in G$; Then,

  $$a \sim b \iff a \circ b^{-1} \in H$$

  where $\circ$ could indicate group operations $+$, or $\cdot$ for which the inverses of b are $-b$ and $\frac{1}{b}$ respectively. In our case, we have the two groups

  $$\mathbb{Q}^\times = \{q \in Q : q \neq 0\}$$

  and

  $$\mathbb{Q}^{\times 2} = \{q^2 : q \in \mathbb{Q}^\times\}$$

  where both are multiplicative. Applying the general definition of quotient groups to these, we have that

  $$\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$$

  contains elements $[x] = x\mathbb{Q}^{\times 2}$ for nonzero $x \in \mathbb{Q}^\times$. Two elements $a, b \in \mathbb{Q}^\times$ are said to be equivalent or otherwise $a \sim b$ if $\frac{a}{b} \in \mathbb{Q}^{\times 2}$ i.e., if $\frac{a}{b}$ is a rational square.

- How are the images of α_E and $\alpha_{E'}$ shown to be finite?

  Resolved: The answer to this is hidden somewhere in the section "*two descent via a 2-isogeny*". We know that the set $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ is infinite – Although this is infinite, $\mathrm{Im}(\alpha_E)$ and $\mathrm{Im}(\alpha_{E'})$ are both finite: For any rational point $P(x, y)$, the x-coordinate can be written as

  $$x = b_1 \left(\frac{M}{e}\right)^2$$

  for some squarefree integer b_1. Rewriting it in this form allows us to show that $x \sim b_1$ as $\frac{x}{b_1} = \left(\frac{M}{e}\right)^2$ is a rational square which belongs to $\mathbb{Q}^{\times 2}$. Then,

  $$\alpha_E(P) = [x] = [b_1]$$

  Substituting this form of x into the equation of the elliptic curve defined by $E : y^2 = x^3 + ax^2 + bx$ with $b = b_1 b_2$ (both $b_1, b_2 \in \mathbb{Z}$) gives us

  $$N^2 = b_1 M^4 + aM^2 e^2 + b_2 e^4$$

  where the new variable N comes from the substitution $y = \frac{b_1 MN}{e^3}$.
  For any element $[x] = [b_1] \in \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$, the descent construction gives a factorization $b = b_1 b_2$, i.e., b_1 is a factor of the fixed integer b. Since there are only finitely many factors of b, the set of classes $[b_1]$ must be finite. Hence,

  $$\mathrm{Im}(\alpha_E) < \infty$$

  Similar argument can be applied to show that $\mathrm{Im}(\alpha_{E'})$ is also finite.

- What's the point of considering that diophantine equation?
  Resolved: The answer to this has been answered in the previous question.

- How is this result established: $2^{r+2} \geq f(n)$ from the family of curves $E_n : y^2 = x^3 + nx$ where n is squarefree
  Resolved: By Tate's result,
  $$2^{r+2} = |\text{Im}(\alpha_E)||\text{Im}(\alpha_{E'})|$$
  Thus, it is enough to show that $f(n) \leq \text{Im}(\alpha_E)$ from which it follows that $f(n) \leq 2^{r+2}$ since $\text{Im}(\alpha_{E'})$ is finite and at least 1. Consider the curve $E : y^2 = x^3 + ax^2 + bx$ with $a = 0$ and $b = n$. If we show that every factorization of n counted by $f(n)$ corresponds to a distinct element of $\text{Im}(\alpha_E)$, we are done.
  Consider a factorization counted by $f(n)$, $n = ab$ such that n is square-free. Then, a and b must also be square-free and $a + b = m^2$ for some $m \in \mathbb{Z}$. Considering the quartic diophantine equation defined earlier, set $b_1 = a$, $b_2 = b$, and $M = e = 1$ to get
  $$N^2 = a + b$$
  Thus, $N = m$ is a solution of the descent equation. Precisely, $(N, M, e) = (m, 1, 1)$ is the solution of the diophantine equation. Since there's a solution to the diophantine equation, the previously established result shows that
  $$x = a \qquad y = ma$$
  is a solution of the equation defining the elliptic curve $E_n : y^2 = x^3 + nx$ since $(am)^2 = a^2m^2 = a^2(a + b) = a^3 + a^2b = a^3 + a(ab) = a^3 + an$. Therefore,
  $$\alpha_{E_n}(P) = [x] = [a]$$
  Each factorization of $f(n)$ corresponds to different elements in $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ precisely because of the condition that n must be square-free. Since n is square-free, a, a' correspond to different classes if they are distinct positive divisors of n. That is, $[a] \neq [a']$. Thus,
  $$f(n) \leq |\text{Im}(\alpha_E)|$$

- What does this notation mean: $O(n^\epsilon)$
  Resolved: If $f(n) = O(n^\epsilon)$; Then, for a fixed $\epsilon > 0$ and sufficiently large n
  $$|f(n)| \leq Cn^\epsilon$$
  for a constant $C = C(\epsilon)$.

- What's a Dirichlet series?
  Resolved: A Dirichlet series is a series of the form
  $$S(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$
  where s is a complex number and a_n is a sequence of complex numbers. The Riemann-zeta function $\zeta(s)$ is a standard example of a Dirichlet series with $a_n = 1$ for $n \geq 1$.

- What does this notation mean: $O_\epsilon(B(x))$
  Resolved: Same as $O(n^\epsilon)$, where instead of n^ϵ we have $B(x)$; The only difference being that the dependence of the constant on ϵ is made explicit here.

- What's the Mobius function?
  Resolved: The Mobius function is a function that essentially detects if a number is square-free. It also captures the parity of the number of distinct prime factors of the number n. It is defined by
  $$\mu(n) = \begin{cases} 1 & n = 1 \\ 0 & \text{n is divisible by a square of a prime} \\ (-1)^k & \text{n is square-free and has } k \text{ distinct prime factors} \end{cases}$$

This crops up in our field as it helps us determine whether a number n is square-free or not:

$$\mu^2(n) = \sum_{d^2|n} \mu(d) = \begin{cases} 1 & \text{n is square-free} \\ 0 & \text{n is not square-free} \end{cases}$$

- What's the Tauberian theorem?
  Resolved: Lemma 1 is the classical form of Tauberian theorem – in general, we are able to deduce some asymptoptic information about a particular sequence/series from information about a related transform, such as a Dirichlet series, provided certain other conditions are satisfied. To put it vaguely, it helps us to transform analytic information to information about the original sequence/series.

- What's the Gauss sum form of the Legendre symbol?
  Resolved: The Gauss sum refers to (here) the sum:

$$\tau = \sum_{b=1}^{p-1} \left(\frac{b}{p}\right) e\left(\frac{b}{p}\right)$$

  where $e(t) = e^{2\pi it}$. Using the fact that $\left(\frac{b}{p}\right) \in \{1, -1\}$, τ can be rewritten as the sum of the summation of $e(b/p)$ over all nonzero quadratic residues and the summation of $-e(b/p)$ over all quadratic non-residues as follows:

$$\tau = \sum_{\text{quadratic residues}} e(b/p) - \sum_{\text{quadratic non-residues}} e(b/p)$$

- What does $R(x; a, q)$ mean exactly?
  Resolved: For our case, we consider when $\gcd(a, q) = 1$. Consider the sequence

$$a, a+q, a+2q, \ldots a+nq$$

  $R(x; a, q)$ is precisely the number of squarefree integers in this progression that are at most x.

6 Things I learned from this paper

Other than what's already been resolved, I have learnt a few strategies/techniques involved in the proofs of some of the lemmas and theorems.

- Squarefree sieve method using Mobius function.
- how changing the order/structure of a sum can expose an estimate
- how the Dirichlet series encodes an arithmetic sequence, and then reveal asymptotic equivalence for the summatory function of the arithmetic sequence.

7 What remains unclear

All the black boxes mentioned in the document:

- Proofs of Lemmas 1 (tauberian theorem), and 2 (squarefree AP estimate)
- Some of the algebraic manipulations in the proof of Lemma 5
- Full descent argument; not covered in the paper, but briefly mentioned in Section 2.
- Proofs of estimates

$$R(x) = \frac{x}{\zeta(2)} + O(\sqrt{x}), \qquad \sum_{\substack{n \le x \\ \text{n squarefree}}} \frac{1}{n} = \frac{\log x}{\zeta(2)} + O(1), \text{ and } \sum_{\substack{n \le x \\ \text{n squarefree}}} \frac{1}{\sqrt{n}} = \frac{2\sqrt{x}}{\zeta(2)} + O(\log x)$$

- How did the Gauss sum equation come into place? How is the relation between that and Legendre symbol established?
- Proof for the claim: For the group $E(\mathbb{Q})$, the additive inverse of a point $P(a, b)$ is given by $-P = -(a, b) = (a, -b)$
- Projective geometry and results from those aren't immediately obvious to me.
- I've only scratched the surface of Algebraic Geometry in an attempt to learn about elliptic curves.

8 Observations and connections

- Connection between Mordell-Weil rank, and rank of a matrix.
- Considering the Gauss sum form of Legendre Formula to deal with the more familiar exponential function; exploiting the fact that $e^{a+b} = e^a e^b$. This strategy can be generalized to show that it might sometimes be worthwhile to rewrite an expression/function as a sum of basic functions such as e^x.
- What stood out to me was how the authors repeatedly manipulated the sums – by splitting, swapping, making it a double sum (by fixing one and varying others in the inner sum; for example, check out the first step in the proof for Lemma 4);
- The unbounded nature of Mordell-Weil rank and the function $f(n)$ has been established in Section 2 via the inequality $f(n) \leq 2^{r+2}$
- Exploiting multiplicative nature of Mobius function, to simplify the estimates.

9 Things I particularly liked

- I liked how every discrete function sum involving discrete number-theoretic functions like $\sum_{n \leq x} \mu^2(n) d(n)$, was approximated by a continuous expression like $Cx \log x$.
- Case-by-case analysis while deriving the final $S(x)$ estimate.
- The connection across fields; in particular, the connection between Analytical NT and Algebraic Geometry through the Mordell-Weil rank of elliptic curves, felt fascinating to me.
- The descent method itself, to show that $\mathrm{Im}(\alpha_E)$ and $\mathrm{Im}(\alpha_{E'})$ is finite by considering the factorizations of $b = b_1 b_2$.